

Aurick Anwar

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EDUCATION

McMaster University

Bachelor of Engineering: Mechatronics Engineering

Hamilton, Canada

September 2025 – June 2030

WORK EXPERIENCE

Software Engineering Intern

HermesAI

Toronto, Ontario

May 2026 - Present

Founding Engineer

Magnified Systems

Toronto, Ontario

February 2026 - Present

- Developing a **mounted device** that can be attached to a helmet and is used to detect and quantify **impact severity** in real time.
- Built and trained a PyTorch regression model that predicted an impact severity score from (1-100) at a **96.4%** accuracy off sample crash data (CSV file).
- Processed a **6-axis IMU** sensor and, using **C++**, developed a severity score algorithm based off linear and angular acceleration of the impact.

Assistant Instructor

The STEAM Project

Toronto, Ontario

June 2025 – August 2025

- Guided **~10** students per session in hands-on woodworking and robotics projects, safely supervising tools such as scroll saws with **zero safety incidents**.
- Co-developed and improved a Battle Bots curriculum, teaching CAD design and basic electronic platforms to **50+ students** improving project completion rates by **~40%**.
- Assisted in packaging and delivering **100+** project kits across multiple locations, ensuring timely distribution.

CMO and Co-Founder

Orbitview

Toronto, Ontario

December 2023 – June 2025

- Led a group of **15-20** young high school developers to build a digital platform for more professional content creation.
- Created documentation and guides for AI resume templates and fixes.
- Planned technical events and collaborated with multiple AI startups in Toronto to host webinars. Advertised on all social media platforms by creating posts to attract over **200** people to our webinar, and our revenue exceeded the initial cost by **5x**.

PERSONAL PROJECTS

Autonomous Self-Driving Vehicle with CARLA, CARLA, YOLOv11, PyTorch

- Developed an end-to-end autonomous driving pipeline in the **CARLA** simulator integrating **YOLO-based object detection** and **vehicle control**, achieving a real-time simulation at **~20-30 FPS**.
- Built and wrote to a dataset with 5+ features (steer, brake, speed, acceleration, throttle), generating **25k+** sample datasets to train a **PyTorch** model.
- Trained a Neural Network, using **Cross Entropy Loss**, achieving **~88.5%** prediction directional accuracy for precise automation navigation.

AI Smart Computer w/ Hand Gesture Control, MediaPipe, PyAutoGUI, OpenCV

- Implemented **6+ gesture mappings** with 21-point hand tracking, enabling click, cursor, mute, and volume control via distance thresholds.
- Reduced unintended inputs by **~60%** using gesture filtering and cooldown logic, achieving **<100 ms** latency.

Basketball Shot Predictor, OpenCV, YOLOv11, PyTorch

- Built a real-time basketball shot prediction system, processing **~30 FPS** video and tracking ball positions across **50+ frames** per shot.
- Created and wrote to a dataset with **9+ features** per frame (speed, distance to rim, position of ball) generating **100+** labeled data points across multiple videos for training.
- Designed and trained a PyTorch neural network, using **BCE loss**, achieving **~80-85%** prediction accuracy on if the shot goes in.

Technical Skills

- Programming Skills: Python, C++
- Libraries: PyTorch, OpenCV, MediaPipe, PyAutoGUI, YOLOv11
- 3D Design: AutoCAD, Fusion360, Inventor
- Electronics and Microcontrollers: Arduino, Raspberry PI, ESP32
- Microsoft Office: Excel, PowerPoint, Word